ALLEN

(Academic Session: 2026 – 2027)

TALLENTEX PRACTICE QUESTIONS

CLASS 10

BIOLOGY

Your Hard Work Leads to Strong Foundation

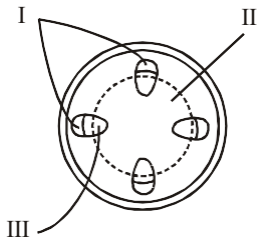
Practice Questions

1. Which pairing is correct?

(1) Sericulture – Fish (2) Pisciculture – Birds

(3) Apiculture – Honeybee (4) Aquaculture – Mosquito

2. The diagram below shows a transverse section of a dicotyledon stem. Which part(s) is/are involved in carrying water and minerals up the stem?



(1) I only (2) II only (3) III only (4) I and III only

3. Suppose you were a neuroscientist and had been given a sample of a new snake venom. You test its effect on action at a synapse, and find that it increases the magnitude of the normal depolarizing excitatory response. The most likely explanation for this is that the venom is –

(1) Blocking release of the neurotransmitter from the vesicles

(2) Binding with the neurotransmitter receptors to interfere with neurotransmitter binding

(3) Binding with the neurotransmitter receptors to mimic the action of the neurotransmitter

(4) Acting to break – down the neurotransmitter in the synaptic cleft

4. Which of the following is not true for excretion in plants?

(1) In plants no definite excretory system or organ is present for removal of the wastes.

(2) In higher plants the waste materials are deposited in various body parts.

(3) CO_2 and O_2 are produced as wastes during respiration by plants.

(4) Tannin, essential oils, latex, gums are examples of plant waste products.

5. Match the columns and select the correct option.

	Column – A		Column – B
(A)	Tendon	(p)	Fluid connective tissue
(B)	Blood	(q)	Blood clotting
(C)	WBC	(r)	Connects muscle to bone
(D)	Thrombocytes	(s)	Protection against disease causing microbes

(1) A-s; B-p; C-q; D-r (2) A-r; B-p; C-s; D-q

(3) A-p; B-q; C-r; D-s (4) A-r; B-p; C-q; D-s

6. When a person breathes in, what happens to the diaphragm and to the ribs?

	Diaphragm	Ribs
(1)	Becomes flat	Moves downwards and inwards
(2)	Becomes flat	Moves upwards and outwards
(3)	Becomes more curved	Moves upwards and outwards
(4)	Becomes more curved	Moves downwards and inwards

7. A biologist studied the concentration of urine produced by a terrestrial mammal, a freshwater fish and a marine fish. Which row of observations would be the most likely for these organisms in their natural environment?

	Terrestrial mammal	Freshwater fish	Marine fish
(1)	Produces dilute urine	Produces concentrated urine	Produces dilute urine
(2)	Produces concentrated urine	Produces dilute urine	Produces dilute urine
(3)	Produces dilute urine	Produces concentrated urine	Produces concentrated urine
(4)	Produces concentrated urine	Produces dilute urine	Produces concentrated urine

8. The table shows the speed at which nerve impulses travel in axons (nerve fibers) of different diameters.

Axon Type	Diameter (mm)	Speed (m/s)
A	13 – 20	80 – 120
B	6 – 12	35 – 75
C	1 – 5	5 – 35
D	0.2 – 1.5	0.5 – 2.0

A reasonable hypothesis based on these data is that as the diameter of an axon increases –

(1) the sensitivity of the nerve decreases

(2) the sensitivity of the nerve increases

(3) the speed of nerve transmission decreases

(4) the speed of nerve transmission increases

9. Given is a table describing the social organization of Honey bees. Which of the following is a correct match of the description of honeybees?

	Types of Honey Bees	Description
I.	 Drone	Female honey bees which prepare honey.
II.	 Queen	Fertile female which lays eggs.
III.	 Worker	Sterile males which look after the young ones, collect nectar and pollen.

(1) I and II are correct (2) I and III are correct

(3) Only II is correct (4) II and III are correct

10. Compare the movement of substances in xylem and phloem tissues.

Feature	Xylem	Phloem
Material transported	Water and minerals	I
Process of movement	II	Translocation
Name of relevant theory	III, IV	Mass flow theory

Replace I, II, III and IV with the appropriate words.

(1) I – sucrose, II – ascent of sap, III – root pressure theory, IV – guttation

(2) I – starch, II – ascent of sap, III – guttation, IV – transpiration pull

(3) I – sucrose, II – ascent of sap, III – root pressure theory, IV – transpiration pull

(4) I – glucose, II – pressure flow hypothesis, III – root pressure theory, IV – adhesion cohesion theory

11. The anterior pituitary gland facilitates growth of an individual by release of the human growth hormone (HGH) which, in turn is regulated by two hormones namely growth hormone releasing hormone (GHRH) and growth hormone inhibiting hormone (GHIH). Imbalance of these hormones could result in gigantism (an individual gains excessive height), dwarfism (a short statured individual) or acromegaly (thickening of limbs, fingers and toes). Interpret the data given below and select the appropriate statement.

Individual	Age group	Hormones
1	2 – 5 yrs.	Excessive GHRH
2	2 – 5 yrs.	Normal GHRH
3	30 – 35 yrs.	Excessive GHRH
4	30 – 35 yrs.	Excessive GHIH
5	2 – 5 yrs.	Excessive GHIH

(1) 1 and 3 will lead to gigantism while 4 and 5 will show dwarfism

(2) 3 will show gigantism, 1 will show acromegaly and 4 and 5 will show dwarfism

(3) 2, 3 and 4 will show normal growth

(4) 1 will show gigantism, 3 will show acromegaly and 5 will show dwarfism

12. The principal nitrogenous excretory compound in humans is synthesized

(1) in liver but eliminated mostly through kidneys.

(2) in kidneys but eliminated mostly through liver.

(3) in kidneys as well as eliminated by kidneys.

(4) in liver and also eliminated by the same through bile.

13. Which one of the following statements is incorrect?

(1) The medullary zone of kidney is divided into a few conical masses called medullary pyramids projecting into the calyces.

(2) Inside the kidney the cortical region extends in between the medullary pyramids as renal pelvis.

(3) Glomerulus along with Bowman's capsule is called the renal corpuscle.

(4) Renal corpuscles, proximal convoluted tubule (PCT) and distal convoluted tubule (DCT) of the nephron are situated in the cortical region of kidney.

14. Two test tubes are filled with a solution of bromothymol blue. A student exhales through a straw into each tube, and the bromothymol blue turns yellow. An aquatic green plant is placed in each tube, and the tubes are corked. One tube is placed in the dark, and the other tube is placed in direct sunlight. The yellow solution in the tube in sunlight turns blue, while the one in the dark remains yellow. Which statement best explains why the solution in the tube placed in sunlight returns to a blue colour?

(1) Oxygen was produced by photosynthesis.
(2) Oxygen was removed by respiration.
(3) Carbon dioxide was removed by photosynthesis.
(4) Carbon dioxide was produced by respiration.

15. Identify the correct names of hormones from the following hints.

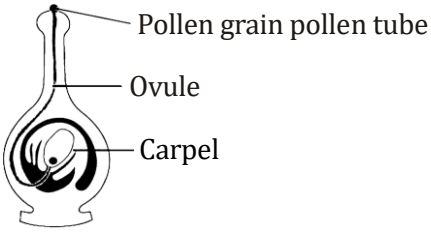
I. "X" hormone induces fruit ripening.
II. "Y" hormone promotes root initiation and also play a role in callus differentiation
III. "Z" hormone increases the tolerance of plant to various stresses and play an important role in seed dormancy.

(1) Y = ABA; X = Auxin; Z = GA
(2) Z = GA; X = Auxin; Y = Ethylene
(3) Y = Auxin; X = Ethylene; Z = GA
(4) X = Ethylene; Y = Auxin; Z = ABA

16. Which of the following diagrams shows hydrotropism?

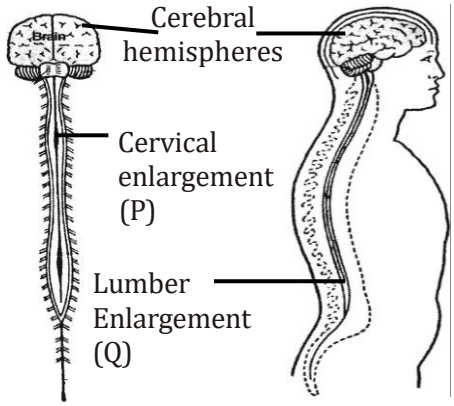
(1) 

(2) 

(3) 

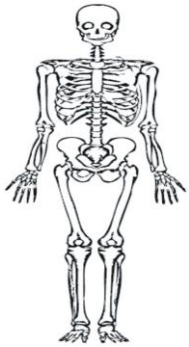
(4) 

17. Look at the diagram given below and identify the position from where nerves to the upper limb originate.

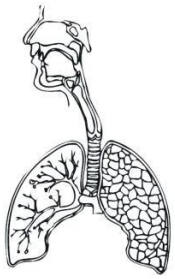


(1) P (2) Q (3) Both P and Q (4) None of these

18. Which of the following is human excretory system?

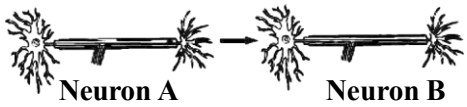
(1) 

(2) 

(3) 

(4) 

19. The below picture shows two neurons. A nerve impulse moves from Neuron A to Neuron B because of –



(1) the release of various carbohydrates into the synapse.

(2) the release of chemicals called neurotransmitters.

(3) the dendrites that are connected.

(4) Neuron A hitting against Neuron B.

20. Match the terms given under Column 'A' with their functions given under Column 'B' and select the answer from the options given below.

Column – A		Column – B	
(A)	Lymphatic System	(i)	Carries oxygenated blood
(B)	Pulmonary vein	(ii)	Immune response
(C)	Thrombocytes	(iii)	To drain back the tissue fluid to the circulatory system
(D)	Lymphocytes	(iv)	Coagulation of blood

(1) (A) – (ii), (B) – (i), (C) – (iii), (D) – (iv)

(2) (A) – (iii), (B) – (i), (C) – (iv), (D) – (ii)

(3) (A) – (i), (B) – (iii), (C) – (ii), (D) – (iv)

(4) (A) – (ii), (B) – (i), (C) – (iii), (D) – (iv)

21. Feeling the tremors of an earthquake, a scared person of seventh floor of a multistoreyed building starts climbing down the stairs rapidly. Which hormone initiates this action?

(1) Insulin (2) Adrenaline (3) Glucagon (4) Gastrin

22. Which of the following diagrams shows the flow of air in our body?

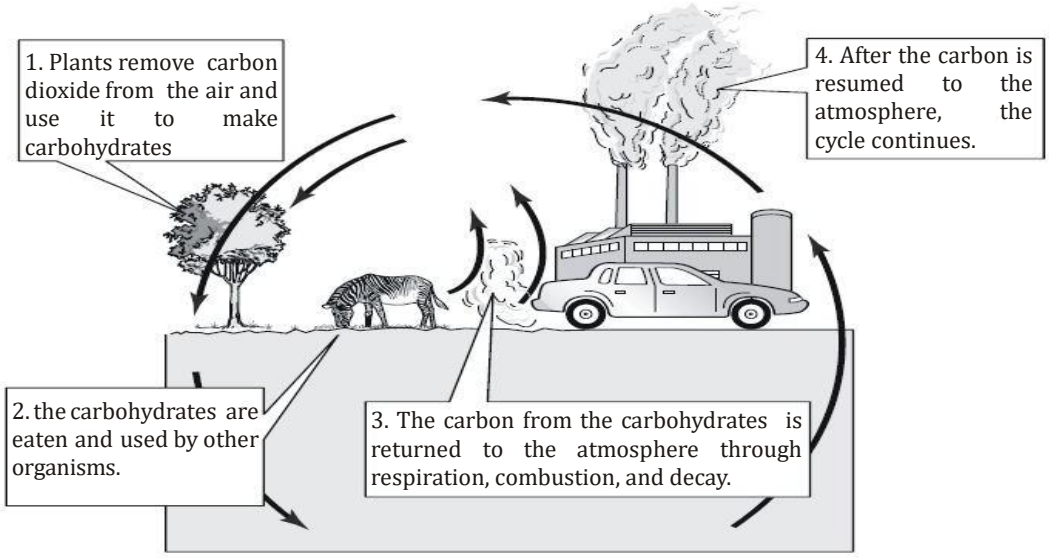
(1) Lungs, Heart, Windpipe, Nose

(2) Lungs, Windpipe, Windpipe, Nose

(3) Nose, Heart, Windpipe, Lungs

(4) Nose, Lungs, Windpipe

23. In the given diagram, at which step, light energy converted into chemical energy?



(1) Step – 1 (2) Step – 2 (3) Step – 3 (4) Step – 4

24. How does alveolar pressure compare to atmospheric pressure, during inspiration?

(1) Alveolar pressure is greater than atmospheric pressure.

(2) Alveolar pressure is less than atmospheric pressure.

(3) Alveolar pressure is the same as atmospheric pressure.

(4) Abdominal pressure is lower than atmospheric pressure.

25. The rhythmic pattern of breathing is established by –

(1) cerebrum (2) cerebellum

(3) reflex – action (4) medulla oblongata

26. Which of these statements about hypothalamic – releasing hormones is true?

(1) They are secreted into blood capillaries.

(2) They are transported by portal veins to the anterior pituitary.

(3) They stimulate the secretion of specific hormones from the anterior pituitary.

(4) All of the above are true.

27. Which of the following is not the function of kidney?

(1) Ultrafiltration (2) Reabsorption

(3) Secretion (4) Urea formation

28. The main function of loop of Henle is the reabsorption of water from the contents of renal tubules. In which of the following will the loop of Henle be poorly developed or even absent?
(1) Desert mouse (2) Carnivorous bird
(3) Freshwater fish (4) Desert Lizard

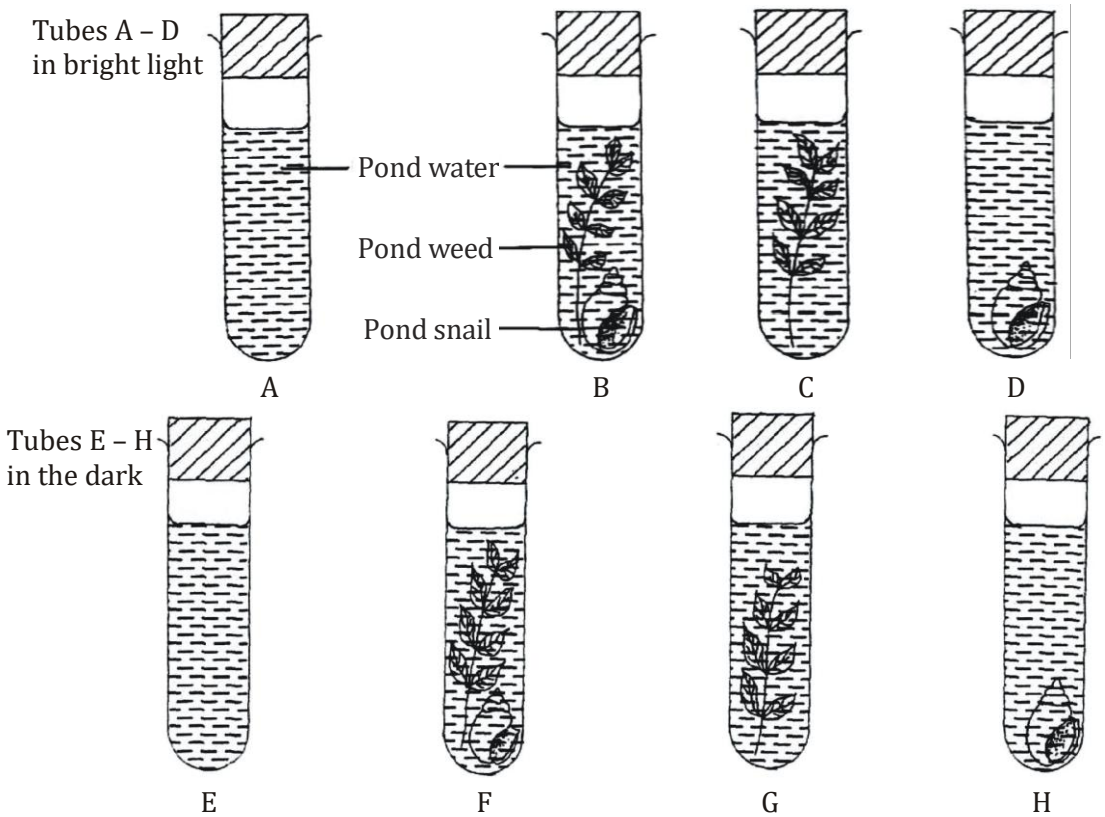
29. In normal human which of the following is completely reabsorbed by Nephrons?
(1) Glucose (2) Protein (3) Water (4) Sodium

30. Which of the following is mismatched?
(1) Medulla oblongata → Regulate involuntary activities
(2) Hypothalamus → Appetite & satiety centre, osmoregulation, voluntary movements
(3) Cerebellum → Equilibrium & posture of body
(4) Spinal cord → Butterfly shaped gray matter

31. The cerebrum wraps around a structure called thalamus, which is –
(1) A major coordinating centre for sensory signal only
(2) A major centre for motor signalling
(3) A major coordinating centre for sensory and motor signalling
(4) Not a nervous part of brain

32. Which of the following substances need to be eliminated from the body?
(a) Glycogen (b) Fats (c) Urea (d) Carbon dioxide
(1) a and d (2) Only c (3) c and d (4) a and c

33. Test tubes A – D are placed in light for 24 hours; Tubes E – H are placed in darkness for 24 hours. Which of the tubes should have the –
a. highest concentration of oxygen after 24 hours?
b. highest concentration of carbon dioxide after 24 hours?



(1) (a) – E, (b) – D (2) (a) – A, (b) – H (3) (a) – G, (b) – B (4) (a) – C, (b) – H

34. Which one of the following is produced in the kidneys?
(1) Rennin (2) Renin (3) ADH (4) Aldosterone

35. Blood calcium level is a resultant of how much dietary calcium is absorbed, how much calcium is lost in the urine, how much calcium is released from bones into the blood and how much calcium from blood enters tissues. A number of factors play an important role in these processes. Mark the one which has no role.
(1) Vitamin – D (2) Parathyroid hormone
(3) Thyrocalcitonin (4) Thymosin

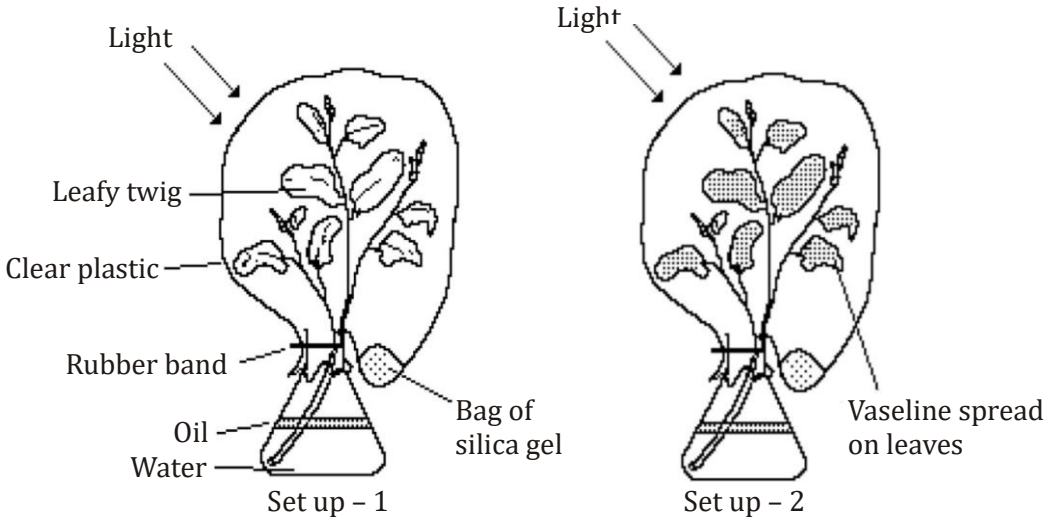
36. In an axon of a neuron, nerve impulse travels –
(1) Away from cyton (2) Towards cyton
(3) Both away and towards cyton (4) Not known

37. The usefulness of fermentation as a means of deriving energy is limited because –
(1) it cannot generate enough ATP
(2) it produces too much NH_2
(3) the end products are non – toxic to the producer
(4) it uses more energy than it produces

38. (i) Cerebellum has very convoluted surface in order to provide the additional space for more neurons.
(ii) The medulla is connected to the spinal cord.
(iii) Medulla contains cardiac and respiratory centres. Which of the above statement/s is/are correct?
(1) All are correct (2) Only (i) is correct
(3) Only (i) and (iii) are correct (4) Only (ii) is correct

39. Insects have a structure analogous to the mammalian kidney called Malpighian tubules, which remove metabolic nitrogenous wastes from the insect's haemolymph. What sort of nitrogenous waste would the Malpighian tubules be removing?
(1) $C_6H_{12}O_6$ (2) Uric acid (3) CO_2 (4) Amino acid

40. In an investigation of factors affecting water loss by plants, the following two experimental setups were prepared, and were left for two hours. Conditions in the setups were similar, except for the factors noted.



Silica gel absorbs water. The silica gel was weighed before and after the two – hour period and the increase in weight was taken.

Comparison of the mass of the silica gel from setups 1 and 2 would help determine if

(1) gas exchange is necessary for photosynthesis

(2) light is necessary for photosynthesis

(3) water is lost via the leaves

(4) stomatal opening is dependent on the CO_2 concentration within the leaf

41. You need pears for a large party after three days but they are not ripe enough to use. What is the best way to hasten the ripening process?

(1) To place the pears in the dark

(2) To place the pears in a refrigerator

(3) To place the pears on the window sill

(4) To place the pears in brown paper bags together with ripe apples

42. Pulmonary artery carries deoxygenated blood from –

(1) right ventricle (2) right atrium

(3) left atrium (4) left ventricle

43. Totally submerged aquatic plants can cause a pH change in the surrounding water when they carry out photosynthesis. What pH change happens and what causes it?

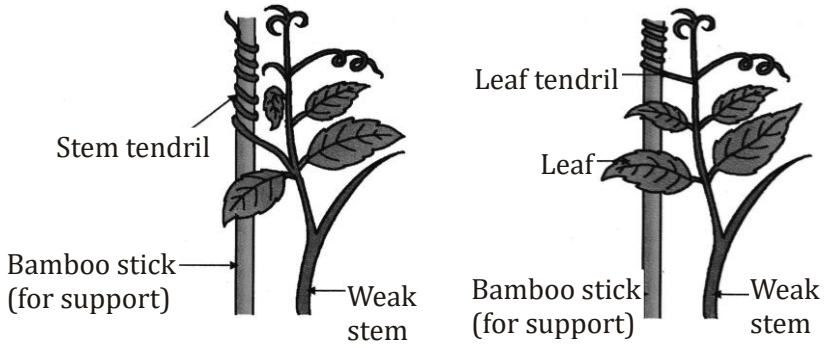
(1) The pH falls because carbon dioxide is absorbed.

(2) The pH rises because carbon dioxide is absorbed.

(3) The pH falls because oxygen is released.

(4) The pH rises because oxygen is released.

44.



Which of the following statement is correct about above figure.

(i) Stem tendril of grapevine and gourd show this type of movement.

(ii) Leaf tendrils of pea show this type of movement.

(iii) Rapid cell division takes place in tendrillar cells that are in contact with support.

(iv) Rapid cell division takes place in tendrillar cells that are away from support.

(1) (i) and (iii) (2) (i), (ii) and (iv)

(3) (ii) and (iii) (4) (ii) and (iv)

45. In which of the following maximum reabsorption of glomerular filtrate occurs?

(1) Proximal convoluted tubule (2) Distal convoluted tubule

(3) Henle's loop (4) Collecting duct

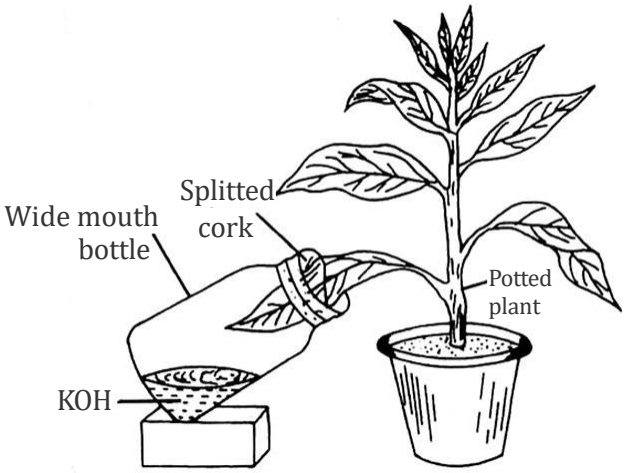
46. In which of the following plants, there will be no transpiration?
(1) Completely submerged aquatic plants
(2) Plants growing in deserts
(3) Plants in tropical rainforest
(4) Plants growing in hilly regions

47. Which statement is not true about thyroxine?
(1) It regulates carbohydrates, protein and fat metabolism in the body.
(2) Thyroid gland requires iodine to synthesize thyroxine.
(3) Thyroxine hormone regulates blood calcium levels.
(4) Thyroxine is responsible for metamorphosis in frog.

48. CO_2 is carried by haemoglobin as
(1) Carboxyhaemoglobin (2) Carbaminohaemoglobin
(3) Oxyhaemoglobin (4) Both (1) and (2)

49. Where is larynx positioned in the respiratory tract?
(1) Between pharynx and trachea (2) Between bronchi and alveoli
(3) Between trachea and bronchi (4) Between pharynx and nasal cavity

50. The given demonstration of an experiment is to prove that



(1) Sunlight is necessary for photosynthesis
(2) Chlorophyll is essential for photosynthesis
(3) O_2 is evolved during photosynthesis
(4) CO_2 is necessary for photosynthesis

51. Dialysis fluid contains all the constituents as in plasma except
(1) Electrolytes (2) Carbohydrate
(3) Nitrogenous wastes (4) All of above

52. Ascent of sap by transpiration and root pressure cause water to rise in plant by
(1) Pulling it upward (2) Pulling and pushing respectively
(3) Pushing it upward (4) Pushing and pulling respectively

53. Which of the following hormone is released from pituitary?
(1) Thyroxine (2) Adrenaline (3) Insulin (4) Growth hormone

54. Double circulation is found in –
(1) turtle (2) dog fish (3) snake (4) both (1) and (3)

55. Which tissue possesses the following characteristics?
• Responsible for growth in plants
• Thin cell wall
• Large nucleus
• Vacuoles absent or very small
(1) Parenchyma (2) Collenchyma
(3) Meristematic tissue (4) Sclerenchyma

56. Which epithelial tissue is chiefly involved in diffusion and is found lining the alveoli and blood capillaries?
(1) Cuboidal epithelium (2) Ciliated epithelium
(3) Squamous epithelium (4) Glandular epithelium

57. Cardiac muscles are characterized by
(1) Voluntary, spindle-shaped cells (2) Involuntary, branched and striated fibres
(3) Voluntary, cylindrical fibres (4) Non-striated spindle-shaped fibres

58. Which of the following tissues provides flexibility to young stems and petioles?
(1) Sclerenchyma (2) Collenchyma (3) Xylem (4) Phloem

59. Organic farming minimizes the use of chemicals. Which of the following organisms are used as biofertilizers?
(1) Anabaena (2) Nostoc (3) Both (1) and (2) (4) Rhizopus

60. Which of the following is correctly matched?
(1) Rabi crop – Gram, Mustard (2) Kharif crop – Wheat, Gram
(3) Agronomy – Production of flowers (4) Olericulture – Production of cereals

61. The time and frequency of irrigation depend upon
(1) Crop (2) Soil (3) Season (4) All of these

62. Which one of the following is a bottom feeder in composite fish culture?
(1) Catla (2) Rohu (3) Mrigal (4) Silver carp

63. The secretion of bile helps in
(1) Digestion of proteins (2) Digestion of starch
(3) Emulsification of fats (4) Digestion of cellulose

64. Which organ is NOT a part of the alimentary canal?
(1) Liver (2) Colon (3) Stomach (4) Small intestine

65. Digestion of proteins begins in
(1) Mouth (2) Stomach (3) Small intestine (4) Large intestine

66. Roughage mainly consists of
(1) Proteins (2) Vitamins (3) Fibres (4) Fats

67. Which gland secretes trypsin, lipase and amylase?
(1) Liver (2) Salivary gland (3) Pancreas (4) Gall bladder

68. The site where gaseous exchange takes place in humans is
(1) Trachea (2) Bronchi (3) Bronchioles (4) Alveoli

69. Respiration involves the following steps:
A. Diffusion of gases across alveolar membrane
B. Transport of gases by blood C. Cellular respiration
D. Pulmonary ventilation E. Diffusion of gases between blood and tissues
The correct sequence is
(1) A → B → C → D → E (2) D → A → B → E → C
(3) E → D → C → B → A (4) C → B → E → A → D

70. The part of respiratory system where gaseous exchange actually occurs is
(1) Bronchi (2) Bronchioles (3) Alveoli (4) Trachea

71. During inhalation,
(1) Diaphragm becomes dome-shaped and ribs move downward.
(2) Diaphragm flattens and ribs move upward and outward.
(3) Air pressure inside lungs becomes greater than atmospheric pressure.
(4) Lung volume decreases.

72. During vigorous exercise, heartbeat increases mainly because
(1) Body temperature decreases (2) Muscles require more oxygen and nutrients
(3) Blood pressure falls suddenly (4) Digestion becomes faster

73. The material transported through phloem is mainly
(1) Water (2) Minerals (3) Sucrose (4) Oxygen

74. Which tissue is mainly responsible for upward transport of water and minerals in plants?
(1) Phloem (2) Xylem (3) Collenchyma (4) Parenchyma

75. Which blood vessel carries oxygenated blood from lungs to the heart?
(1) Pulmonary artery (2) Aorta
(3) Pulmonary vein (4) Vena cava

76. Which of the following chambers of the human heart has the thickest muscular wall?
(1) Right atrium (2) Left atrium (3) Right ventricle (4) Left ventricle

77. The principal nitrogenous waste excreted by humans is
(1) Uric acid (2) Ammonia (3) Urea (4) Creatinine

78. Maximum reabsorption of glucose, amino acids and salts occurs in the
(1) Bowman's capsule (2) Loop of Henle
(3) Proximal convoluted tubule (4) Collecting duct

79. Glomerulus along with Bowman's capsule is collectively called
(1) Renal pelvis (2) Renal corpuscle
(3) Ureter (4) Nephron loop

80. Which hormone regulates the level of glucose in blood?
(1) Thyroxine (2) Insulin (3) Adrenaline (4) Growth hormone

ANSWER KEY

Que.	**1**	**2**	**3**	**4**	**5**	**6**	**7**	**8**	**9**	**10**	**11**	**12**	**13**	**14**	**15**
Ans.	3	3	3	3	1	2	4	4	3	3	4	1	2	3	4
Que.	**16**	**17**	**18**	**19**	**20**	**21**	**22**	**23**	**24**	**25**	**26**	**27**	**28**	**29**	**30**
Ans.	4	1	4	2	2	2	2	1	2	4	4	4	3	1	2
Que.	**31**	**32**	**33**	**34**	**35**	**36**	**37**	**38**	**39**	**40**	**41**	**42**	**43**	**44**	**45**
Ans.	3	3	4	2	4	1	1	1	2	3	4	1	2	2	1
Que.	**46**	**47**	**48**	**49**	**50**	**51**	**52**	**53**	**54**	**55**	**56**	**57**	**58**	**59**	**60**
Ans.	1	2	3	1	4	3	2	4	4	3	3	2	2	3	1
Que.	**61**	**62**	**63**	**64**	**65**	**66**	**67**	**68**	**6 9**	**7 0**	**71**	**72**	**73**	**74**	**75**
Ans.	4	3	3	1	2	3	3	4	2	3	2	2	3	2	3
Que.	**76**	**77**	**78**	**79**	**80**										
Ans.	4	3	3	2	2										